LLM Foundations Homework Sheet

Required preparation for Module 0: LLM Basics

Name ______________________ **Date** ______________________

Required workload: Watch the 8-minute 3Blue1Brown warm-up and the full 3h31m Karpathy video outside class. You may watch Karpathy at 1.25x or 1.5x, but pause at each checkpoint and answer in your own words. The class session will use this worksheet.

Required Videos

1. Warm-Up Video

3Blue1Brown: Large Language Models explained briefly

https://www.youtube.com/watch?v=LPZh9BOjkQs

Purpose: simple visual model of next-token prediction, chatbots, pretraining, and transformers.

2. Main Homework Video

Andrej Karpathy: Deep Dive into LLMs like ChatGPT

https://www.youtube.com/watch?v=7xTGNNLPyMI

Purpose: full foundation for data, tokens, training, inference, hallucinations, tool use, RLHF, and practical use.

Before Watching

In one sentence, what do you think ChatGPT is doing when it answers a question?

Tiny Glossary

Term	Meaning for this course
Token	A chunk of text the model processes. It can be a word, part of a word, punctuation, or code fragment.
	The current working memory: the text the model can see while answering.

Context window	
Pretraining	Learning broad patterns from massive text and code data.
SFT	Supervised fine-tuning: training from example conversations or desired outputs.
RLHF	Reinforcement learning from human feedback: using human preferences to shape model behavior.
Inference	Running the trained model to generate output, usually one token at a time.
Hallucination	A fluent answer that is unsupported, misleading, or false.
Tool	An external action the system lets the model request, such as search, calculation, or code execution.
Guardrail	A rule, approval step, or check that limits risky behavior.

3Blue1Brown Warm-Up Reflection

After the short video, write two ideas that became clearer.

What is one question you still have before watching Karpathy?

Karpathy Checkpoints

Timestamps are based on the current YouTube chapters for "Deep Dive into LLMs like ChatGPT." Use them as anchors, not as exact quiz boundaries.

`00:01:00` **Pretraining data (internet)**

What kinds of data help train an LLM? Why does data quality matter?

00:07:47 Tokenization

What is a token? Give one example where tokenization might surprise a user.

00:14:27 Neural network input/output

What goes into the model, and what comes out?

00:20:11 Neural network internals

You do not need the math. What is the useful mental model for what the network has learned?

00:26:01 Inference

Why does the answer appear one piece at a time?

00:31:09 GPT-2: training and inference

What does a base model do? How is that different from a helpful chat assistant?

00:42:52 Llama 3.1 base model inference

What did you notice about interacting with a base model?

00:59:23 Pretraining to post-training

Why is pretraining not enough if we want a useful assistant?

01:01:06 Post-training data (conversations)

How do conversation examples change the behavior of a model?

01:20:32 Hallucinations, tool use, knowledge, working memory

Why can a model sound confident and still be wrong? How can tools help?

01:41:46 Knowledge of self

Why might a model be confused about what it is, what version it is, or what it can access?

01:46:56 **Models need tokens to think**

Why can writing intermediate reasoning or scratch work help a model perform better?

02:01:11 **Tokenization revisited**

Why can spelling, counting letters, or manipulating strings be surprisingly hard?

02:04:53 **Jagged intelligence**

What does it mean for a model to be brilliant at one task and weak at another?

02:07:28 **SFT to reinforcement learning**

What changes when the system moves beyond examples and starts using feedback or rewards?

02:14:42 **Reinforcement learning**

Explain reinforcement learning in one sentence using your own words.

02:48:26 **RLHF**

Why do human preferences matter when building assistant-like models?

03:21:46 **Grand summary**

Write three takeaways you want to remember before learning agents.

Bridge to Agents

Use this course definition:

Practical agent = LLM + goal + context + tools + loop + guardrails

Classify each example as **chatbot answer**, **tool call**, or **agent loop**.

Example	Your classification	Why?
ChatGPT explains the difference between pretraining and post-training.		
A model asks a calculator tool to compute 438 * 27.		
A coding system reads files, proposes a plan, edits code, runs tests, and asks before deploying.		

Verification Habit

Pick one important claim from the video or from a chatbot answer. How would you verify it?

Exit Ticket

In 4-6 sentences, explain why an agent needs guardrails.

Completion Checklist

- [ ] I watched the 3Blue1Brown warm-up video.
- [ ] I watched the full Karpathy video, using the checkpoint questions.
- [ ] I completed the glossary and checkpoint responses in my own words.
- [ ] I can explain the difference between chatbot answer, tool call, and agent loop.
- [ ] I wrote an exit ticket about guardrails.

Teacher Grading Guide

Criteria	Points
Completed video checkpoints with specific, non-copy-pasted responses	40
Accurate use of core vocabulary: token, context, pretraining, post-training, inference, hallucination	20
Clear explanation of hallucination and verification behavior	15
Correctly distinguishes chatbot answer, tool call, and agent loop	15
Exit ticket explains why guardrails matter for risky actions	10